

UHMWPE Natural

Direc

PE-1000

(Ultra High Molecular Weight Polyethylene)

Colour: Natural

Density: 0.930g/cm³**Materi**

Mechanical Properties	Value	
Tensile strength at yield	17	
Elongation at yield	11	
Modulus of elasticity in tension	700	
Modulus of elasticity in flexure	800	
Impact strength	No br.	
Creep rupture strength after 1000hrs with static load	-	
Time yield limit for 1% elongation after 1000hrs	-	
Coefficient of friction against hardened and ground steel	0.29	
P= 0.05 N/mm ² , v= 0.6m/s		
Ball indentation hardness	35	
Shore hardness D	64	

Thermal Properties	Value	
Glass transition temperature	-	
Crystalline melting point	135	
Service temperature (short term)	120	
Service temperature (long term)	90	
Coefficient of thermal expansion	20	
Specific heat	1.84	
Coefficient of thermal conductivity	0.41	V

Electrical properties	Value	
Surface resistivity	>10 ¹⁴	
Dielectric strength	44	

Other properties	Value	
Water absorption	<0.01	
Resistance to hot water, washing soda	Resistant	
Resistance to weathering	No	
Flammability (UL94)	HB	